

## Hermetically-Sealed Bending Beam

### FEATURES

- Capacities: 10, 20, 30, 50, 75, 100, 200 and 250 kg
- Stainless steel or alloy steel construction
- Stainless steel version hermetically-sealed
- High side load tolerance
- Easy installation
- OIML C3 approval from 50 kg to 250 kg

### APPLICATIONS

- Platform scales (multiple load cells)
- Silo/hopper/tank weighing
- Packaging machines
- Dosing/filling
- Belt scales/conveyor scales

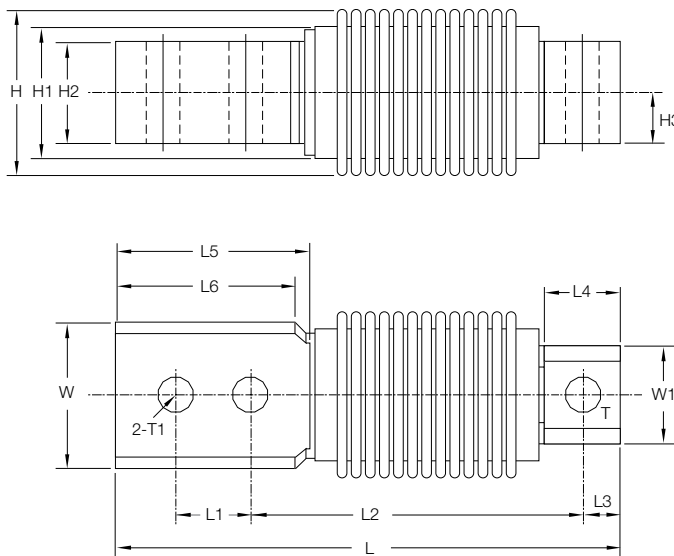
### DESCRIPTION

The Model HBB is a single-ended bending beam load cell designed for multiple cell applications, such as low profile platform scales or small tank scales, when used with proper mounting accessories. It is insensitive to side load and capable of reversed loading.

The Model HBB is constructed of stainless steel or alloy steel. The stainless steel version is hermetically-sealed to IP68, providing excellent protection against corrosive and wash-down environments.



### OUTLINE DIMENSIONS



#### Wiring Diagram

#### Stainless Steel:

|              |        |
|--------------|--------|
| + Excitation | Red    |
| - Excitation | Black  |
| + Signal     | Green  |
| - Signal     | White  |
| + Sense      | Brown  |
| - Sense      | Yellow |

#### Alloy Steel:

|              |       |
|--------------|-------|
| + Excitation | Red   |
| - Excitation | Black |
| + Signal     | Green |
| - Signal     | White |

| Stainless Steel Capacities (kg)                           |      | L    | L1   | L2   | L3   | L4   | L5   | L6  | H    | H1   | H2   | H3   | W    | W1   | T         | T1     |
|-----------------------------------------------------------|------|------|------|------|------|------|------|-----|------|------|------|------|------|------|-----------|--------|
| 10, 20, 50, 100, 200*, 250                                | mm   | 120  | 18   | 82   | 10   | 19   | 41   | 38  | 41.6 | 28.2 | 20   | 10   | 31.5 | 26.6 | Ø 8.2     | Ø 8.2  |
|                                                           | inch | 4.72 | 0.71 | 3.23 | 0.39 | 0.75 | 1.61 | 1.5 | 1.64 | 1.11 | 0.79 | 0.39 | 1.24 | 1.05 | Ø 0.32    | Ø 0.32 |
| * Threaded hole version "200KGT" available: T = M10 × 1.5 |      |      |      |      |      |      |      |     |      |      |      |      |      |      |           |        |
| Alloy Steel Capacities (kg)                               |      | L    | L1   | L2   | L3   | L4   | L5   | L6  | H    | H1   | H2   | H3   | W    | W1   | T         | T1     |
| 10, 20, 30, 50, 75, 100, 200, 250                         | mm   | 120  | 18   | 82   | 10   | 19   | 41   | 38  | 41.6 | 31.5 | 20   | 10   | 31.5 | 26.6 | M10 × 1.5 | Ø 8.2  |
|                                                           | inch | 4.72 | 0.71 | 3.23 | 0.39 | 0.75 | 1.61 | 1.5 | 1.64 | 1.24 | 0.79 | 0.39 | 1.24 | 1.05 | M10 × 1.5 | Ø 0.32 |

## Hermetically-Sealed Bending Beam

| SPECIFICATIONS                              |                                       |                                   |                       |
|---------------------------------------------|---------------------------------------|-----------------------------------|-----------------------|
| PARAMETER                                   | VALUE                                 |                                   | UNIT                  |
| NTEP/OIML Accuracy class                    | Non-Approved                          | C3 (stainless steel version only) |                       |
| Maximum no. of intervals (n)                | 1000                                  | 3000*                             |                       |
| $Y = E_{\max}/V_{\min}$                     | 5000                                  | 10000                             | Maximum available     |
| Standard capacities ( $E_{\max}$ )          | 10, 20, 30**, 50, 75**, 100, 200, 250 |                                   | kg                    |
| Rated output—RO                             | 2.0                                   |                                   | mV/V                  |
| Rated output tolerance                      | 0.25                                  |                                   | ±% of rated output    |
| Zero balance                                | 1                                     |                                   | ±% of rated output    |
| Non linearity                               | 0.030                                 | 0.025                             | ±% of rated output    |
| Hysteresis                                  | 0.030                                 | 0.025                             | ±% of rated output    |
| Non-repeatability                           | 0.020                                 |                                   | ±% of rated output    |
| Creep error (20 minutes)                    | 0.030                                 | 0.020                             | ±% of rated output    |
| Zero return (20 minutes)                    | 0.030                                 | 0.020                             | ±% of rated output    |
| Temperature effect on min. dead load output | 0.0026                                | 0.0014                            | ±% of rated output/°C |
| Temperature effect on sensitivity           | 0.0015                                | 0.0010                            | ±% of rated output/°C |
| Compensated temperature range               | -10 to +40                            |                                   | °C                    |
| Operating temperature range                 | -20 to +60                            |                                   | °C                    |
| Safe overload                               | 150                                   |                                   | % of RC               |
| Ultimate overload                           | 300                                   |                                   | % of RC               |
| Excitation, recommended                     | 10                                    |                                   | VDC or VAC RMS        |
| Excitation, maximum                         | 15                                    |                                   | VDC or VAC RMS        |
| Input impedance                             | 385±5                                 |                                   | Ω                     |
| Output impedance                            | 350±3                                 |                                   | Ω                     |
| Insulation resistance                       | >5000                                 |                                   | MΩ                    |
| Cable length                                | 3                                     |                                   | m                     |
| Construction                                | Stainless steel or alloy steel        |                                   |                       |
| Environmental protection                    | IP68 (stainless steel version only)   |                                   |                       |

\* Capacities: 50–250 kg

\*\* Capacities of 30 kg and 75 kg as alloy steel version only

All specifications listed subject to change without notice.

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